Flood & Debris Situation Report · White Soil Pocket Area

Analysis e231d2afced4

Location **White Soil Pocket Area (27.82°N, 85.04°E), Nepal** Event **26 Aug 2026** Area **207.4 km²** Issued **14 Oct 2026** Baseline OSM **27 Jul 2026** Grid **10 m**

140 buildings in high-confidence damage · 2 bridges likely lost (high confidence) · 4 communities cut off

1. KEY FIGURES

12.23 km²
Area hit
0.36 km² new water · 0.03 km² not observable

4
Cut off
+0 without hospital access

8
Bridges lost
of 51 checked · 7 at risk

586
Buildings hit
of 21,149 mapped · 619 near cut-off communities

44
Longer route only
longest extra time 0.5 h

WHAT CHANGED
clear-sky optical, 73.2% of area

RIVER WIDTH
115 → **250 m** **+135 m**
12 points · 10 widened ⩾ 50 m

RIVER MOVED SIDEWAYS
170 m at most
median 38 m

OPEN WATER
2.40 → **2.12 km²** **−12%**
water surface seen from space

BARE CHANNEL
3.39 → **5.08 km²** **+50%**
water, mud and fresh sediment, no plants

2. SITUATION

Where did the flood hit?

Comparing images from before and after 26 Aug 2026, we estimate **12.23 km²** of debris and scour and **0.36 km²** of new water in 207.4 km². Terrain hid 0.03 km² from the radar: unknown, not safe.

How we estimate: radar change between the before and after passes, stronger than normal pass-to-pass change, backed by optical plant loss where clear; low ground beside the mapped flow and below its level is filled in at low confidence.

Who is cut off?

4 communities are likely cut off by road and **0** more likely cannot reach a hospital (0 of 74 health facilities likely isolated). **44** communities likely need a longer route, up to 0.5 h extra.

How we estimate: routes on the pre-event road map are recomputed without impassable sections; cut off means no route to a town or hospital.

What was damaged?

586 mapped buildings are likely hit. About **16.4 km** of road and trail (227 segments) is likely impassable. Of 51 mapped bridges, **8** are likely lost and **7** possibly at risk.

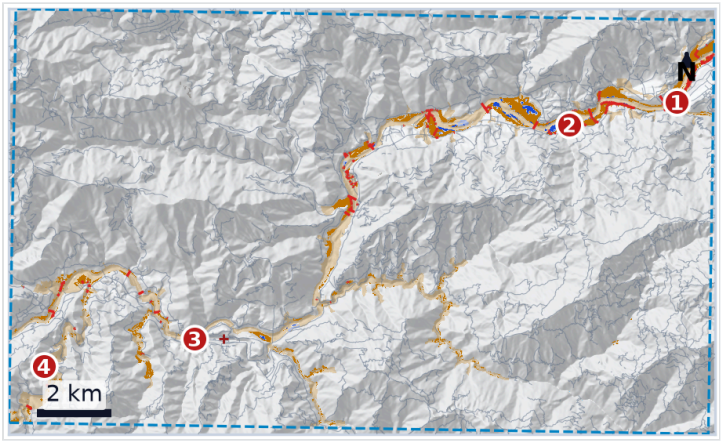
How we estimate: a building, road or bridge counts as hit when it overlaps mapped flood or debris; bridges by radar change over the span.

What to reopen first?

Reopening **Road Pasang Lhamu Highway** (primary) first would likely restore road access to **1** community and **124** buildings, more than any other single section.

How we estimate: each impassable section is ranked by how many communities and buildings regain a route if it alone reopens.

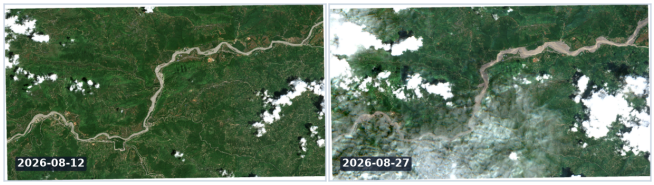
MAP 1 3. SITUATION MAP



Debris & scour · New water · Impassable road · Cut off · No hospital access · Not observable

Boundaries and names shown do not imply official endorsement. Map data: OpenStreetMap (pre-event); images: Copernicus Sentinel-1/2.

MAP 2 4. BEFORE AND AFTER



Sentinel-2 true colour · 12 Aug 2026 (before) → 27 Aug 2026 (after)

12.23 km²	0.36 km²	0.00 km²	0.03 km²
DEBRIS & SCOUR	NEW WATER	WATER BEFORE	NOT OBSERVABLE

5. CUT-OFF COMMUNITIES · TOP 4 OF 4

Settlement	Access status	Bldgs	Pre dist (km)	Maps
1 देवीघाट (hamlet)	Cut off	310	5.9	Map
2 **(unnamed)** (hamlet)	Cut off	124	8.9	Map
3 महेश-दोमान (hamlet)	Cut off	123	0.9	Map
4 **(unnamed)** (village)	Cut off	62	8.9	Map

6. BRIDGES · 8 LOST · 7 AT RISK

Bridge / highway	Status	Radar drop (dB)	Span (m)	Maps
Unnamed (footway)	Likely lost	6.7	151	Map
Belkhu Khola Bridge	Likely lost	7.8	84	Map
devighat-Tadi bridge	Likely lost	2.8	61	Map

7. REOPEN FIRST

#	Asset / roadway	Restores villages	Restores bldgs
1	Road Pasang Lhamu Highway (prima...	1	124
2	Road (service)	1	123

DATA

Radar: Sentinel-1 R085 ascending (16 Aug 2026 → 28 Aug 2026) · R019 descending (24 Aug 2026 → 5 Sept 2026)
Optical: Sentinel-2 12 Aug 2026 → 27 Aug 2026
DEM & OSM: Copernicus DEM GLO-30 · OSM 27 Jul 2026
Grid: 10 m, local UTM

LIMITATIONS

- Images are days apart: damage between them is mapped together, and nothing is forecast.
- Ground hidden from the radar by steep terrain is unknown, not undamaged.
- Optical evidence only where the sky was clear; cloud elsewhere.
- 10 m pixels: a building counts as hit when its footprint overlaps mapped change; damage to a single house is not graded.
- Low-confidence areas (including valley floor filled in from the observed flood level) are inferred, not seen; severity uses high-confidence evidence only.
- Roads, bridges and buildings missing from the pre-event map are not assessed; routes assume every mapped road was open before.
- Buildings are counted, not people: no population data is used, so people displaced is not measured.
- Not field-verified. Confirm on the ground before acting.

LINKS

provenance.json (scenes & parameters)
download.zip (all GIS layers)
Interactive map dashboard

Credits: Contains modified Copernicus Sentinel data 2026. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors (ODbL), via the ohsome API. Training data: Kuro Siwo (Bountos et al., NeurIPS 2024).